CHAPTER 13

Conditional probabilities and binomial distribution

S36 (reminders), S37, S38, S39, S41

Before you begin · Conditional probability

Without a calculator, in six minutes. Answer briefly. Use trees and events. Then choose the targeted review indicated under the questions.

1. Express " A given B" .
2. Write $P(A \cup B)$ when A and B are mutually exclusive.
3. Calculate $P(A \cap B)$ if $P(A) = 0.4$ and $P_A(B) = 0{,}3$.
4. Give the complement of the event "at least 3 successes".
5. What does the independence of A and B ?
6. Give the expectation of a fair die.

Answers

$P_B(A)$; $P(A) + P(B)$; 0.12 ; at most 2 successes ; $P(A \cap B) = P(A)P(B)$; 3,5.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Name the reference population after "given". **2.** Take the union of two disjoint cases. **3.** Multiply the probability of the branch by the conditional probability; R46. **4.** List the integers less than 3; R48. **5.** Go back to factorising the probability of the intersection. **6.** Add up the six weighted values by a sixth. The R questions serve as follow-up material with the teacher, even if their collective routine is planned for later.

Objectives

Build a tree; choose a binomial distribution; calculate a cumulative probability and a threshold; interpret expectation and variance.

Course notes

Reminders: conditioning and independence

If $P(A) > 0$, $P_A(B) = \frac{P(A \cap B)}{P(A)}$. A tree branch carries a conditional probability; the probability of a path is the product of its branches. For a partition $A_1, \ldots, A_r$, $P(B) = \sum_i P(A_i \cap B)$; if all the $P(A_i) > 0$, it can be written $\sum_i P(A_i) P_{A_i}(B)$. Two events are independent when $P(A \cap B) = P(A)P(B)$. For $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. Do not confuse $P_A(B)$ and $P_B(A)$.

Example. $P(A) = 0{,}4$, $P_A(B) = 0{,}7$, $P_{\overline{A}}(B) = 0{,}2$. Then $P(B) = 0{,}4 \times 0{,}7 + 0{,}6 \times 0{,}2 = 0{,}4$ and $P_B(A) = \frac{0{,}28}{0{,}4} = 0{,}7$.

Bernoulli and binomial

A Bernoulli trial has two outcomes, success with probability p and failure with probability $1 - p$. The indicator variable B is 1 for a success and 0 otherwise: $E(B) = p$, $V(B) = p(1-p)$. A Bernoulli scheme is a repetition of n identical and independent trials. The number X of successes follows the binomial distribution $\mathcal{B}(n, p)$.

Distribution and parameters

For $k \in \{0, \dots, n\}$,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

$$E(X) = np, \qquad V(X) = np(1-p), \qquad \sigma(X) = \sqrt{np(1-p)}.$$

Proof of the distribution. Each path comprising k successes has prob. $p^k(1-p)^{n-k}$. There are $\binom{n}{k}$ choices for the positions of the successes. These paths are mutually exclusive; add their probabilities.

Cumulative events

$$P(X \leq k) = \sum_{j=0}^{k} P(X = j), \qquad P(X \geq k) = 1 - P(X \leq k-1).$$

$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a-1).$$

For at least one success, $P(X \geq 1) = 1 - (1-p)^n$. The complement of "at least k" is "strictly fewer than k".

Example. For $X \sim \mathcal{B}(10, 0{,}2)$, $E(X) = 2$ and $V(X) = 1{,}6$. $P(X = 2) = 45(0{,}2)^2(0{,}8)^8 \simeq 0{,}3020$ and $P(X \geq 1) = 1 - 0{,}8^{10} \simeq 0{,}8926$. The expectation is neither a certain outcome nor necessarily the most likely.

Threshold and interval

To look for the smallest n such that $1 - (1-p)^n \geq q$, a loop over n is enough. Using logarithms, we can solve $(1-p)^n \leq 1 - q$ then round in the right direction. To find the smallest integer k such that $P(X > k) \leq \alpha$, increase k until obtaining $P(X \leq k) \geq 1 - \alpha$, then check the previous index. We can also look for an interval $[a, b]$ containing at least a prescribed probability, without assuming that it is unique.

Independent trials, with more than two outcomes

For independent experiments, the probability of a list of results is the product of the probabilities of its components. The sample space is the Cartesian product of the sample spaces of each experiment. By example, for two independent rolls of a fair die, the sample space contains the 36 pairs (i, j), $1 \leq i, j \leq 6$, each with probability $\frac{1}{36}$. The probability that the sum is 5 equals $\frac{4}{36} = \frac{1}{9}$, as the favourable pairs are $(1,4), (2,3), (3,2), (4,1)$. In a Bernoulli scheme, success is coded by 1 and failure by 0: the sample space is $\{0,1\}^n$. Independence concerns the whole sequence; a simple equality between two observed probabilities is not enough to prove it.

Build the model before calculating

13.1 From the table to the tree

First use: S36.

Worked example · Changing the denominator

Out of 1,000 objects, 600 come from workshop A and 400 from B. A produces 12 defective objects and B produces 20. Denoting by D a defect, $P(A \cap D) = 12/1000$, $P_A(D) = 12/600 = 0{,}02$ and $P_D(A) = 12/32 = 0.375$. The same count 12 appears, but the reference sample space changes.

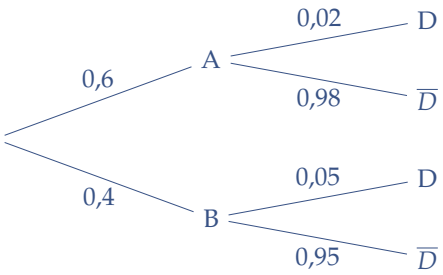


Figure 13.1. Read path A then D: $0.6 \times 0{,}02 = 0.012$, or 12 items out of 1,000.

A1 · Immediate application Complete the table with two rows A/B and two columns D/not D. Calculate $P(D)$ in two ways. Are the defect and the workshop independent?

13.2 Recover a branch and reverse a conditioning

First use: S38.

Property · Partition and the law of total probability

If A and $\overline{A}$ form a partition, $P(D) = P(A)P_A(D) + P(\overline{A})P_{\overline{A}}(D)$, when the conditional probabilities are defined. This result comes from the disjoint union of the two paths leading to D.

Worked example · A missing piece of data

We know $P(A) = 0{,}7$, $P_A(D) = 0{,}1$ and $P(D) = 0{,}13$. Let $x = P_{\overline{A}}(D)$. The equation $0{,}13 = 0{,}7 \times 0{,}1 + 0{,}3x$ gives $x = 0{,}2$. The result belongs to $[0;1]$, which is a necessary check.

A2 · Partial guidance With $P(A) = 0{,}4$, $P_A(D) = 0.2$ and $P(D) = 0.14$, find $P_{\overline{A}}(D)$ then $P_D(A)$.

A3 · Refute "If 90% of patients test positive, then 90% of people who test positive are ill." Build a table for 1,000 people, with 100 ill, 90 positive among the ill and 90 positive among those who are not ill, then calculate the two proportions.

13.3 Four features to check

First use: S39 and S41.

Definition · Bernoulli scheme

We repeat a fixed number n of independent trials, with two outcomes and the same probability of success p. The number X of successes follows the distribution $\mathcal{B}(n;p)$: $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$ for $0\leq k\leq n$.

Worked example · Counting paths

For four trials with probability 1 /2, getting exactly two successes corresponds to six paths, each with probability 1 /16. Thus $P(X=2)=6/16$. Independence allows us to multiply the probabilities along a path; the binomial coefficient counts the placements of the successes.

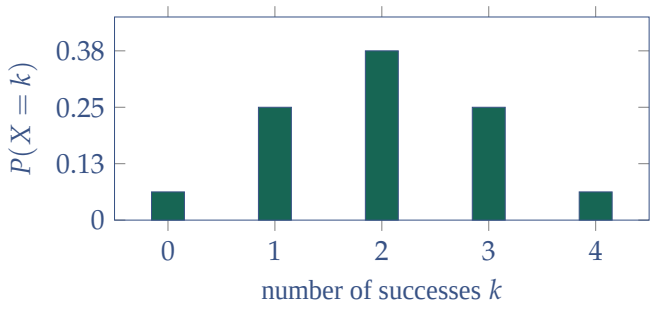


Figure 13.2. Law $\mathcal{B}(4;1/2)$: explain the symmetry and check the sum of the heights.

A4 · Accept or refuse State in which cases the binomial model is justified: ten independent tosses of a coin; ten draws without replacement from an urn containing five red balls and five blue ones ; number of tosses before the first success. Justify each answer.

13.4 Translate words into complete events

First use: S41.

Worked example · A boundary shifted by one unit

For X integer, "strictly more than 4" means $X\geq 5$ and its complement is $X\leq 4$. "At least 4" means $X\geq 4$ and its complement is $X\leq 3$. Finally, $P(2\leq X\leq 5)=P(X\leq 5)-P(X\leq 1)$.

A5 · Immediate application For $X\sim\mathcal{B}(8;0{,}3)$, write in terms of cumulative probabilities the events: at least 3 successes; strictly fewer than 3; between 2 and 5 inclusive. Calculate exactly $P(X=0)$.

13.5 Find a probability rather than a number of trials

First use: S42.

Worked example · Two trials

For two independent trials with probability p, the probability of at least one success is $1-(1-p)^2$. Making it equal to 3/4 amounts to $(1-p)^2 = 1/4$. Since $1-p \geq 0$, we keep $1-p = 1/2$, so p=1/2. The root with the negative sign would be incompatible with p in [0;1].

A6 · Choice of strategy Three trials are independent, with probability of success p. We want the probability of at least two successes to be 0,9. Express this probability in terms of p, prove the uniqueness of a solution in $[0;1]$ and bound it between two consecutive hundredths.

A first step: Count two and three successes separately. For uniqueness, study a function on $[0;1]$.

One minute, spoken aloud. Explain why a probability tree does not imply independence: the probabilities at the second level may depend on the first outcome.

Delayed review · S50 then S77. Solve a threshold $1-(1-p)^n \geq c$ by logarithm, then compare exact probabilities and concentration bounds.

Investigation

Checking components - 40 min.

A component is chosen at random from the output of two workshops. Let A be event "the component comes from the first workshop", $B = \overline{A}$ and D the event "the component is defective". We are given $P(A) = 0{,}6$, $P_A(D) = 0{,}02$ and $P_B(D) = 0{,}05$. Approximate results will be rounded to 10^{-4}.

1. Represent the situation with a weighted tree, then calculate $P(D)$.

2. The chosen component is defective. Calculate the probability that it comes from the first workshop.

3. Twenty components are now chosen independently, each following the previous model. Let X be the number of defective components among the twenty. Justify the distribution of X and specify its parameters.

4. Calculate $P(X=0)$, $P(X \geq 2)$ and $E(X)$.

5. If the twenty components were taken from the same batch, which hypothesis of the model would need to be re-examined? Explain why.

Graded exercises

E1 · Direct application. Consider two events A and B such that $P(A) = 0{,}3$, $P_A(B) = 0{,}8$ and $P_{\overline{A}}(B) = 0.1$. Calculate $P(B)$, then $P_B(A)$.

E2 · Application and justification. The random variable X follows the binomial distribution with parameters 8 and $\frac{1}{4}$. Give the exact values of $P(X=3)$, $E(X)$ and $V(X)$.

E3 · Method to organise. Let X be a random variable taking integer values. For every integer k, set $F(k) = P(X \leq k)$. Express $P(X \geq 4)$ and $P(2 \leq X \leq 5)$ using F.

E4 · Choice of strategy. Five cards are drawn one after another, without replacement, from a 52-card pack containing four aces. Let X be the number of aces. Does the variable X follow a binomial distribution? Justify.

E5 · Synthesis or proof. We independently repeat a trial whose probability of success is $\frac{1}{2}$. Find the smallest number of trials for which the probability of obtaining at least one success is greater than or equal to 0.99.

Work from the textbook or a past paper

Baccalauréat, Metropolitan France, 17 June 2025: exercise 1, question 5 (p. 3). In class after S41. Read the context of questions 1 to 4; for this work, assume the result $p = 0{,}0714$ from question 4. The paper is available here: official past paper . Use the data and instructions of this past paper.

Homework

DM13 - Bookings and overbooking - 60 min.

A room has eighteen seats. The organiser accepts twenty bookings, paid in advance. Each person who has booked turns up with probability 0,8, independently of the others. Let X be the number of people who turn up. If more than eighteen people come, the extra people are turned away.

1. Justify the distribution of X, then calculate its expectation and variance.

2. Calculate the probability that the organiser must turn away at least one person. Round to 10^{-6}.

3. Each booking earns 10 monetary units, even if the person does not come. Each person turned away receives a total payment of 30 monetary units, refund and compensation included. Let $R = \max(X - 18; 0)$ be the number of people turned away. Express the gain G in terms of R, then calculate $E(G)$.

4. Calculate the gain if the organiser accepts only eighteen bookings, with the same payment rules. Compare the two strategies in terms of mean gain.

5. Name one model assumption that may be debatable in practice, then a criterion other than the monetary gain to be taken into account in this decision.

MCQ A · conditioning and independence

After S37.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

1. $P_A(B)$ denotes:

A. $P(A \cap B)$ B. $P(B \mid A)$

C. $P(A \mid B)$ D. $P(\overline{B})$

2. If $P(A) = 0.4$ and $P_A(B) = 0.5$, then $P(A \cap B)$ is:

A. 0,9 B. 0,2

C. 0,5 D. 0,1

3. Independent events satisfy:

A. $P(A \cap B) = 0$ B. $P(A \cap B) = P(A) + P(B)$

C. $P(A \cap B) = P(A)P(B)$ D. $A = B$

4. "At most 3 successes" translates as:

A. $X < 3$ B. $X \leq 3$

C. $X \geq 3$ D. $X > 3$

5. In a weighted tree, the probability of a path is:

A. the sum of its branches B. the product of its branches

C. always $\frac{1}{2}$ D. a difference

MCQ B · binomial distribution and threshold

After S41; integer counting variables.

For each question, only one option is correct. Justify your choice orally before checking the solutions.

6. $X \sim \mathcal{B}(10; 0{,}3)$ counts:

A. the successes of ten identical and independent trials B. ten probabilities

C. the failures only D. a continuous mean

7. For $X \sim \mathcal{B}(n; p)$, $E(X)$ is:

A. p^n B. np

C. $n + p$ D. $n(1 - p)$

8. "Strictly more than 4" is equivalent to:

A. $X \geq 4$ B. $X \leq 4$

C. $X \geq 5$ D. $X > 5$

9. The assumptions of the Bernoulli scheme are not all satisfied if:

A. there is a fixed number of trials B. success has the same probability

C. the trials are dependent D. the outcome is success/failure

10. To check a minimal integer threshold in a monotonic problem, we also check:

A. the previous index B. a colour in the tree

C. $P(X = 0)$ only D. the name of X

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $P_A(B) = P_B(A)$ in any situation.

2. If $X \sim \mathcal{B}(n; p)$, then $P(X \geq k) = 1 - P(X \leq k - 1)$.

3. An observed frequency is exactly equal to the theoretical probability.

Review of prior work

I know how to build a tree, choose or refuse the binomial model and translate an integer threshold. **To watch out for:** swapping conditioning and intersection; forgetting to shift an integer bound.

MCQ solutions

1. **B.** $P_A(B)$ is the probability of B given A, for $P(A) > 0$. The intersection and the reverse conditioning are different.
2. **B.** $P(A \cap B) = P(A)P_A(B) = 0{,}4 \times 0{,}5 = 0{,}2$. Successive probabilities are multiplied, not added.
3. **C.** By definition of independence, $P(A \cap B) = P(A)P(B)$. An empty intersection corresponds to mutual exclusivity, not to independence in general.
4. **B.** "At most 3" includes 3: $X \leq 3$. "Less than 3" would exclude it.
5. **B.** The probability of a path is the product of successive conditional probabilities. We then add disjoint paths to obtain a union.
6. **A.** The number of trials is 10, the probability of success 0.3, with independence. The variable counts successes, not probabilities.
7. **B.** A sum of n success indicators with expectation p has expectation np. The mean number of failures would be $n(1-p)$.
8. **C.** X is an integer: $X > 4$ is equivalent to $X \geq 5$. $X > 5$ wrongly excludes the value 5.
9. **C.** Independence is one of the assumptions of the Bernoulli scheme. Without it, we can no longer apply the model automatically, even if distributions may happen to coincide in special cases.
10. **A.** Checking the candidate index and the previous one establishes minimality when the probability is monotonic in the size. A single calculation at the candidate index only proves that it works.

True/false. False: with $P(A) = 1/2$, $P(B) = 1/4$ and $B \subset A$, we have $P_A(B) = 1/2$ and $P_B(A) = 1$. True for X integer and k integer: the two events are complementary. False: a single toss of a fair coin has a frequency of 0 or 1, not 1/2.

Solutions to the applications and guidance

Application 1. Row A: 12 and 588; row B: 20 and 380. $P(D) = 32/1000 = 0{,}032 = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05$. $P_A(D) = 0{,}02 \neq 0.032$: no independence.

Application 2. $0{,}14 = 0{,}4 \times 0{,}2 + 0{,}6x$ gives $x = 0{,}1$. $P_D(A) = 0{,}08/0{,}14 = 4/7$.

Application 3. Ill: 90 positive and 10 negative; not ill: 90 positive and 810 negative.
$P_M(+) = 0.9$, but $P_+(M) = 90/180 = 0{,}5$.

Application 4. The first is suitable if we count one fixed outcome. Without replacement, the probabilities change and the draws are not independent. The number of trials until the first success is not not fixed: it is not a number of successes in a scheme of fixed length.

Application 5. $1 - P(X \leq 2)$; $P(X \leq 2)$; $P(X \leq 5) - P(X \leq 1)$. $P(X = 0) = (7/10)^8 = 0{,}05764801$.

Application 6. $q(p) = 3p^2(1-p) + p^3 = 3p^2 - 2p^3$. $q' = 6p(1-p) > 0$ on $]0;1[$, therefore q is strictly increasing on $[0;1]$. Continuity and $q(0) = 0 < 0{,}9 < q(1) = 1$ give existence and uniqueness. $q(0{,}80) = 0{,}896 < 0{,}9 < 0{,}905418 = q(0{,}81)$.

Consolidation

Have students keep the table next to the tree; ask at each step "among whom?". For the thresholds, have the integers represented on a number line before calculating. In the parametric problem, distinguish the variable X, the integer $n = 3$ and the parameter p.

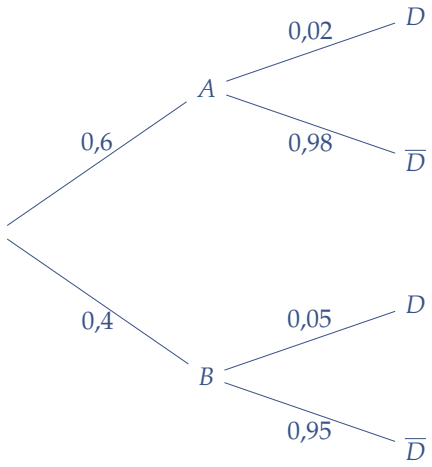
Teaching notes and solutions

Guidance for the teacher

Revisit conditional probabilities in S36 before any Bernoulli scheme. Independence is a model assumption; it does not follow from repetition alone. For drawing without replacement, do not automatically invoke a binomial distribution. The formulas for expectation and variance are used in S39; their proof via a sum of indicator variables is revisited in chapter 14. A threshold required in December must be obtainable with a loop, the logarithm not yet being taught.

Solutions to the activity

1. The tree has the branches A and B, with probabilities 0 ,6 and 0,4. From A, the branches $D, \overline{D}$ carry 0.02, 0.98; from B, they carry 0.05 and 0.95.



By the law of total probability, $P(D) = P(A)P_A(D) + P(B)P_B(D) = 0{,}012 + 0{,}020 = 0{,}032$.

2. $P(D) > 0$ and $P(A \cap D) = 0{,}6 \times 0{,}02 = 0.012$. Therefore $P_D(A) = \frac{P(A \cap D)}{P(D)} = \frac{0{,}012}{0{,}032} = \frac{3}{8} = 0{,}375$.

3. Each sample is a trial with two outcomes, "defective" or "not defective", with the same probability 0.032 of a defect. The twenty samples being independent, $X \sim \mathcal{B}(20, 0{,}032)$.

4. $P(X = 0) = 0{,}968^{20} \simeq 0.5218$. By the complement,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0{,}968^{20} - 20(0{,}032)0{,}968^{19} \simeq 0{,}1332.$$

Finally $E(X) = 20 \times 0{,}032 = 0{,}64$. This expectation does not imply that a sample has a non-integer number of defects.

5. A common cause linked to the batch could make the defects dependent. The binomial distribution and its calculations rely on the independence given in the model; this assumption should be re-examined for another protocol.

E1. By the law of total probability, $P(B) = 0{,}3 \times 0{,}8 + 0{,}7 \times 0{,}1 = 0{,}31$. Since $P(B) > 0$, $P_B(A) = \frac{P(A \cap B)}{P(B)} = \frac{0{,}24}{0{,}31} = \frac{24}{31}$.

E2. By the binomial formula, $P(X = 3) = \binom{8}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^5 = 56\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^5$. The parameters give $E(X) = \frac{8}{4} = 2$ and $V(X) = 8\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \frac{3}{2}$.

E3. The complement of $X \geq 4$ is $X \leq 3$, so $P(X \geq 4) = 1 - F(3)$. For the interval, we remove from $X \leq 5$ the values $X \leq 1$, therefore $P(2 \leq X \leq 5) = F(5) - F(1)$.

E4. No, in a standard pack containing four aces. Without replacement, getting an ace on the first draw changes the probability of getting one on the second: the trials are not independent. Moreover, the number of aces can never be 5, whereas a binomial distribution with parameters 5 and $0 < p < 1$ gives a positive probability to this value.

E5. The probability of at least one success is $1 - (\frac{1}{2})^n$. The condition is equivalent to $2^n \geq 100$. Since $2^6 = 64 < 100 \leq 128 = 2^7$, the first integer is 7. The probability is then $\frac{127}{128} \simeq 0{,}99219$.

Solutions to homework DM13

1. X counts the successes in twenty independent Bernoulli trials with parameter $0,8$. Hence $X \sim \mathcal{B}(20, 0{,}8)$, $E(X) = 20 \times 0{,}8 = 16$ and $V(X) = 20 \times 0{,}8 \times 0{,}2 = 3{,}2$.

2. The room is short of seats if $X > 18$, i.e. $X = 19$ or $X = 20$. Since these events are disjoint,
$$P(X > 18) = 20(0{,}8)^{19}(0{,}2) + (0{,}8)^{20} \simeq 0{,}069175.$$
The risk is about 6.92 %.

3. The number of refusals R equals 0 for $X \leq 18$, 1 for $X = 19$ and 2 for $X = 20$. By definition of the expectation,
$$E(R) = P(X = 19) + 2P(X = 20) \simeq 0{,}0807045.$$
The takings are $20 \times 10 = 200$ and the payments $30R$, therefore $G = 200 - 30R$. Thus $E(G) = 200 - 30E(R) \simeq 197{,}5789$ monetary units. The 30 already include all the payments related to a refusal; 10 must not be subtracted again.

4. With eighteen bookings, no refusal is possible and the takings are $18 \times 10 = 180$, whatever the number of people present. The expected gain with twenty bookings exceeds this amount by about 17.58 monetary units.

5. The arrivals of people from the same group may be dependent. Moreover, the satisfaction of the audience and the consequences of a refusal are not summed up by the expected gain. The result quantifies a model; on its own it is not enough to dictate a decision.

Solutions to the textbook or past-paper work

The 100 choices are independent, with the same probability $p = 0.0714$. Therefore $X \sim \mathcal{B}(100; 0{,}0714)$.

$$P(X \leq 7) = \sum_{k=0}^{7} \binom{100}{k} (0{,}0714)^k (0{,}9286)^{100-k} \simeq 0{,}577.$$

Finally, $E(X) = 7.14$ and $V(X) = 100 \times 0{,}0714 \times 0{,}9286 = 6{,}630204 \simeq 6{,}63$.